

# Control of a Hybrid Thermal Management System: A Heuristic Strategy for Charging and Discharging a Latent Thermal Energy Storage Device

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**Abstract**—The use of a real-time controller for managing the recharging and discharging strategy of the thermal energy storage (TES) device in a hybrid thermal management system (TMS) is critical to realizing the intended performance benefits of such systems. For systems involving rapid cooling of power electronics, such as increasingly electrified air vehicles, new control strategies are needed that can accommodate (1) faster timescales associated with the TES dynamics and (2) limited knowledge of upcoming heat loads. This paper considers a hybrid TMS consisting of a single-phase cooling loop and a phase change material-based TES device. Unlike the model predictive control (MPC) and other optimal methods commonly used to control such systems, this paper implements a heuristic logic-based controller leveraging a higher-order TES model to gain additional knowledge of the internal temperature distribution of the TES. The controller regulates the temperature of the surface of a cold plate attached to a heat source that produces a transient heat load disturbance signal. Additionally, with no knowledge of upcoming disturbances, the controller must manage and conserve the state of charge of the TES device, specifically determining when it is possible to recharge and discharging only when necessary. A simulated case study demonstrates that the controller successfully maintains the cold plate temperature below a critical upper limit of 45°C during a 250 second simulation, with the exception of a 2 second period during the largest heat pulse (6 kW) when the upper limit is exceeded. Furthermore, it is shown that in order for a conventional TMS without TES to maintain the cold plate temperature below the same upper limit, the conventional TMS needs both a heat exchanger and fluid tank volume four times the size of those included in the hybrid TMS. During the same 250 second simulation, the oversized conventional TMS also exceeds the upper limit temperature during the 6 kW heat pulse, albeit for 5 seconds instead of 2; this is due to the slower dynamics of the heat exchanger as compared to those of the TES.

**Index Terms**—Thermal Energy Storage, Phase Change Materials, Thermal Management, Logic-Based Control

## I. INTRODUCTION

With increased electrification in air vehicles, there is a need for thermal management systems (TMSs) that can provide rapid cooling for power electronics generating unpredictable transient heat loads. Hybrid TMSs are those in which the

integration of thermal energy storage (TES) enables heat rejection via a primary heat exchanger as well as temporary absorption of heat via the TES device. Such systems have the potential to provide robustness against highly transient heat loads. However, due to the finite storage capacity of TES devices, hybrid TMSs cannot operate passively; the recharging and discharging cycles of the TES must be managed with a real-time controller.

Demand response controllers for building cooling systems with integrated TES are well-studied and typically leverage the overnight hours for recharging, while discharging occurs during the daytime to maximize economic savings based on variable utility rate structures. These control strategies also often rely on predictions of future heat loads using weather forecasts and estimated cooling demand based on the time of day. On the other hand, controllers for hybrid TMSs in vehicle systems often operate on faster timescales and with substantially greater uncertainty regarding future heat loads. Moreover, the nature of their operation is such that there are not obvious periods of time for TES recharging as in the case of overnight charging of building TES.

Several researchers have studied control of hybrid TMS in the context of building heating and cooling systems in which the objective is to shift the demand to off-peak hours when electricity prices are lower [1]–[6]. These systems often leverage ice or chilled water TES devices to maximize energy storage capacity, and logic-based control techniques for building TES are well documented in the ASHRAE (American Society of Heating, Refrigeration, and Air-Conditioning Engineers) Handbook [7]. Among these techniques is the *chiller-priority* method, which aims to maximize the use of a downsized chiller (the primary heat rejection component) and only use the TES when cooling demands exceed the chiller capability.

Alternatively, some control strategies rely on model predictive control (MPC) or dynamic programming to minimize an objective function [1]–[6], [8]–[10]. However, the use of MPC is challenging for systems with fast dynamics given the computational burden associated with online implementation. Moreover, when controlling the charging or discharging of

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Table I  
LITERATURE REVIEW

|                              | Vehicle TMS | Higher-Order TES Model | Logic-Based Controller | MPC/Optimal Control | No Future Knowledge Assumption |
|------------------------------|-------------|------------------------|------------------------|---------------------|--------------------------------|
| Drees and Braun 1996 [1]     |             |                        | x                      | x                   |                                |
| Henze et al. 1997 [2]        |             |                        | x                      | x                   |                                |
| Henze et al. 2005 [3]        |             |                        |                        | x                   |                                |
| Cole et al. 2012 [4]         |             |                        |                        | x                   |                                |
| Candanedo et al. 2013 [5]    |             |                        | x                      | x                   |                                |
| Shafiei and Alleyne 2015 [8] | x           |                        | x                      | x                   |                                |
| Tam et al. 2019 [6]          |             |                        | x                      | x                   |                                |
| Glos et al. 2020 [9]         | x           |                        |                        | x                   | x                              |
| Pangborn et al. 2020 [10]    | x           |                        |                        | x                   |                                |
| Shanks and Jain 2022         | x           | x                      | x                      |                     | x                              |

a TES device, an estimate of its state of charge (SOC) is required. This can be achieved through estimation of the temperature distribution within the TES device, but accurate estimation requires dynamic TES models comprised of hundreds of state variables that are too complex to be used for real-time control [8]. Instead, TES devices are often modeled as heat exchangers with a single state variable—the state of charge—using an effectiveness-SOC correlation such as that described in [1].

Optimal control approaches work well for controlling TES in buildings, but they often rely on accurate predictions of upcoming heat loads or cooling demands. For example, Candanedo et al. [5] and Henze et al. [3] use weather forecasts and data from previous years to predict the cooling demands over several days or weeks, which are then used as predicted disturbance inputs to their models. Although the integration of TES in a vehicle TMS is well studied [11], literature on control methods for these systems is more sparse. Glos et al. [9] use MPC to control an electric vehicle TMS that combines many subsystems and includes a TES device, but the controller does not specifically use the TES to efficiently reject transient disturbance heat loads. Instead the TES is modeled as a single-state lumped system and used to store thermal energy for cabin heating. Similarly, Pangborn et al. [10] apply MPC to systems with distributed thermal storage but use single-state TES models and assume heat loads are known a priori.

This paper utilizes a simulation model of a single-phase thermal-fluid loop with an integrated phase-change material (PCM)-based TES device. The cooling loop consists of a fluid reservoir, a cold plate, a heat exchanger that can reject heat to a secondary fluid, and a proportional control valve that determines the relative flow rates through the TES device and a bypass line. A heuristic logic-based controller that leverages the additional heat rejection capacity of the TES to regulate the temperature of the cold plate surface is designed and implemented into the simulation model. The controller uses a dynamic model of the TES to predict when recharging and discharging are possible. Temperature feedback from various points in the fluid loop, including the temperature distribution inside the TES, and SOC feedback from the TES, inform the controller's decisions.

This paper is organized as follows. Section II presents a brief description of the models used to simulate the single-phase cooling loop and the integrated TES device. Section III details the calculation of state of charge, the controller objectives, and the proposed heuristic logic-based control strategy. Section IV presents a simulated case study of the hybrid TMS and logic-based controller subjected to randomly varying heat loads to demonstrate the controller's ability to manage multiple objectives. This section then compares the capability of the controlled system to a conventional TMS without integrated TES.

## II. SYSTEM DESCRIPTION

This section outlines the construction of the simulation model for the hybrid thermal management system considered in this paper. The hybrid TMS model builds on previous work by Nash and Jain [12] and Gohil et al. [13]. The authors of [12] derive a first-principles model of a notional single-phase thermal-fluid loop consisting of a fluid tank, a pump, a heat addition component, and a shell-and-tube heat exchanger connected to both the primary fluid loop and a secondary system which provides chilled fluid to the secondary side of the heat exchanger. In this work, the system is simplified by removing the pump model and further modified to include models of a TES device and a cold plate with an integrated electrical resistive heater element as the heat addition component. The heat transfer fluid in the primary and secondary loops is water. The fluid in the tank, the cold plate and heat exchanger surfaces (walls), and the fluid passing through the cold plate and heat exchanger are modeled as lumped masses, each defined using a single state variable. The system diagram of the TMS with the integrated TES is shown in Fig. 1. The following assumptions are made to further simplify the model:

- 1) All tubing has zero length, and transport delays between components are negligible.
- 2) The secondary fluid loop is not modeled. Instead, the chiller fluid temperature,  $T_{ch,f}$ , and the secondary fluid mass flow rate,  $\dot{m}_s$ , are treated as exogenous disturbance inputs which are independent from the primary fluid states.

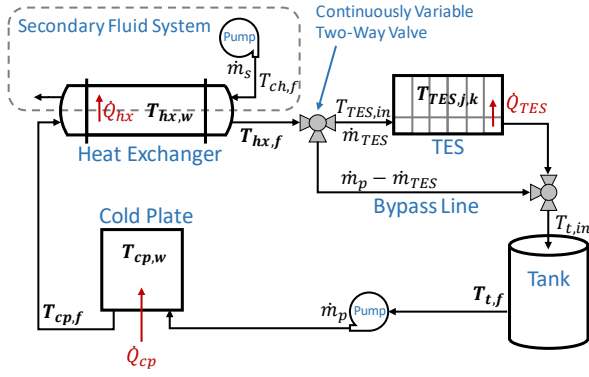


Figure 1. Hybrid TMS diagram. All system states are bolded.

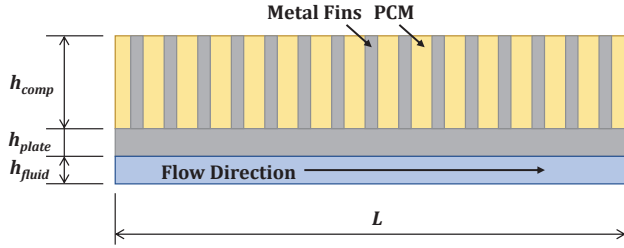


Figure 2. Cross section of the TES device (not to scale). All boundary conditions are adiabatic except for the fluid inlet, which is an input to the model. Not shown in this diagram are rectangular fins in the fluid channel which are parallel to the direction of fluid flow and extend to the full height of the channel.

- 3) The heat transfer fluid is incompressible and has no inertia. Changes in mass flow rate propagate through the fluid loop instantaneously.

A high fidelity dynamic model of a latent TES device was previously derived and validated by Gohil et al. [13]. The device is modeled as a flat plate heat exchanger with PCM embedded between rectangular metal fins. Since the fins are closely spaced (2 mm pitch), the PCM-fin layer is modeled as a uniform composite material, referred to as a composite PCM or CPCM, using the effective medium approximation outlined by Tamraparni et al. [14]. The primary fluid and CPCM are separated by a thin metal plate, and fins in the fluid channel improve the convective heat transfer rate. Fig. 2 illustrates a cross-section of the TES device.

Due to the low thermal conductivity of the CPCM, a lumped model would fail to accurately model the heat diffusion through the device. The proposed heuristic controller, described in more detail in Section III, relies on knowledge of the temperature of the metal separator plate to predict whether the heat transfer rate between the fluid and the CPCM will be positive or negative. For a constant state of charge, depending on the temperature distribution inside the TES, the metal plate temperature could be greater than or less than the fluid temperature; therefore, the heat transfer rate could also be positive or negative. Lumped models that rely on the state of charge to predict the heat transfer rate assume that the entire TES, including the metal plate, is at a uniform

temperature. As such, these models cannot account for the variations in temperature within the TES device caused by rapid transient heat loads. Therefore, both the state of charge and the temperature of the plate must be available to the controller; in this paper, perfect knowledge of all temperature states is assumed.

To create the hybrid TMS model, the TMS and TES models are combined in the configuration shown in Fig. 1. The TES is integrated between the heat exchanger and the tank with a continuously variable two-way valve controlling the relative flow rate through the TES and the bypass line. The primary fluid mass flow rate,  $\dot{m}_p$ , and the mass flow rate of the fluid flowing through the TES device,  $\dot{m}_{TES}$ , are control inputs. The cold plate heater power,  $\dot{Q}_{cp}$ , the secondary fluid mass flow rate,  $\dot{m}_s$ , and the chiller fluid outlet temperature,  $T_{ch,f}$ , are treated as exogenous disturbance inputs. The model is described in more detail in the following subsections.

#### A. Primary Fluid Loop Model

Equations (1)-(4) are the state equations for the cold plate and heat exchanger, as derived in [12] using conservation of energy. The cold plate model has two states: one for the fluid exiting the cold plate,  $T_{cp,f}$ , and one for the cold plate wall,  $T_{cp,w}$ . To model convective heat transfer between the wall and the fluid, the surface temperature is assumed to be uniformly  $T_{cp,w}$ , and the fluid temperature is assumed to be uniformly  $T_{cp,f}$ . The convective heat transfer coefficient is calculated using the empirical power law function  $\alpha_{cp} = K_{cp} \dot{m}_p^{r_{cp}}$ . The convective heat transfer in the primary fluid side of the heat exchanger is modeled using the Gnielinski correlation, and the secondary fluid side is modeled using a correlation for shell-and-tube heat exchangers [15]. As with the cold plate, the heat exchanger has one state for the primary fluid outlet temperature,  $T_{hx,f}$ , and another for the heat exchanger wall temperature,  $T_{hx,w}$ .

$$\dot{m}_{cp,f} c_{p,f} \frac{dT_{cp,f}}{dt} = A_{cp} \alpha_{cp} (T_{cp,w} - T_{cp,f}) + \dot{m}_p c_{p,f} (T_{t,f} - T_{cp,f}) \quad (1)$$

$$\dot{m}_{cp,w} c_{p,cp,w} \frac{dT_{cp,w}}{dt} = A_{cp} \alpha_{cp} (T_{cp,f} - T_{cp,w}) + \dot{Q}_{cp} \quad (2)$$

$$\dot{m}_{hx,f} c_{p,f} \frac{dT_{hx,f}}{dt} = A_{hx,p} \alpha_{hx,p} (T_{hx,w} - T_{hx,f}) + \dot{m}_p c_{p,f} (T_{cp,f} - T_{hx,f}) \quad (3)$$

$$\dot{m}_{hx,w} c_{p,hx,w} \frac{dT_{hx,w}}{dt} = A_{hx,p} \alpha_{hx,p} (T_{hx,f} - T_{hx,w}) - \dot{Q}_{hx} \quad (4)$$

The heat load applied to the cold plate wall,  $\dot{Q}_{cp}$ , is an exogenous disturbance input. The heat transfer rate from the heat exchanger wall to the secondary fluid,  $\dot{Q}_{hx}$ , given in (5), is calculated using the effectiveness-NTU method and is a function of the state variable  $T_{hx,w}$  and the two exogenous

inputs:  $\dot{m}_s$ , the secondary fluid mass flow rate; and  $T_{ch,f}$ , the chiller fluid temperature.  $\dot{Q}_{hx}$  is also treated as a disturbance input to the system. For the cold plate and the primary and secondary sides of the heat exchanger,  $A$  represents the surface area in contact with the fluid and  $\alpha$  represents the convective heat transfer coefficients.

$$\dot{Q}_{hx} = \dot{m}_s c_{p,f} (T_{ch,f} - T_{hx,w}) \left( \exp \left[ -\frac{A_{hx,s} \alpha_{hx,s}}{m_s c_{p,f}} \right] - 1 \right) \quad (5)$$

Downstream of the TES, the tank fluid temperature is determined using an enthalpy balance equation considering mass transfer into and out of the tank as well as the enthalpy stored in the tank fluid. The enthalpy entering the tank due to mass transfer is a function of the temperature of the fluid entering the tank,  $T_{t,in}$ , which is determined by the TES outlet fluid temperature,  $T_{TES,f}$ ; the heat exchanger outlet fluid temperature,  $T_{hx,f}$ ; and the TES and bypass mass flow rates. Likewise, the enthalpy leaving the tank due to mass transfer is a function of the temperature of the fluid leaving the tank and the primary mass flow rate. Assuming perfect mixing within the tank, the temperature of the fluid leaving the tank is equal to the uniform temperature of the fluid within the tank,  $T_{t,f}$ . For (6), the tank fluid temperature state equation, the temperature of the fluid entering the tank is represented as  $T_{t,in}$ . The mass of the fluid in the tank,  $M_{t,f}$ , is constant.

$$M_{t,f} c_p \frac{dT_{t,f}}{dt} = \dot{m}_p c_p (T_{t,in} - T_{t,f}) \quad (6)$$

Equations (1)-(4) and (6) are combined and simplified to form the nonlinear state space model given in (7), where  $\dot{\mathbf{Q}}$ , defined in (8), is the disturbance vector containing the cold plate and heat exchanger heat transfer rates. The state vector  $\mathbf{T}_{fl}$  contains the five primary fluid loop states:  $\mathbf{T}_{fl} = [T_{cp,f} \ T_{cp,w} \ T_{hx,f} \ T_{hx,w} \ T_{t,f}]^T$ . The temperature of the fluid entering the tank,  $T_{t,in}$ , is left as an input in this model so that the TES model can be integrated with the heat exchanger and tank models to form a closed system model of the hybrid TMS.

$$\dot{\mathbf{T}}_{fl} = \mathbf{A}_{fl} \mathbf{T}_{fl} + \mathbf{B}_{fl} \dot{\mathbf{Q}} + \mathbf{B}_t T_{t,in} \quad (7)$$

$$\dot{\mathbf{Q}} = \begin{bmatrix} \dot{Q}_{cp} \\ \dot{Q}_{hx} \end{bmatrix} \quad (8)$$

### B. TES Model

The high-order model of the TES is divided into a grid of  $n_x$  control volumes in the direction parallel to fluid flow and  $n_y + 2$  control volumes in the direction perpendicular to fluid flow, as illustrated by Fig. 3. The CPCM layer is discretized using  $n_x \times n_y$  control volumes, and the fluid channel and metal separator are each discretized into one sublayer of  $n_x$  control volumes. The governing equations, which are presented in detail in [13], are derived using conservation of mass and energy applied to individual control volumes within the fluid channel, metal separator, and CPCM layer.

Equation (9) is the general form of the energy balance for the  $j^{\text{th}}$  control volume along the length of the

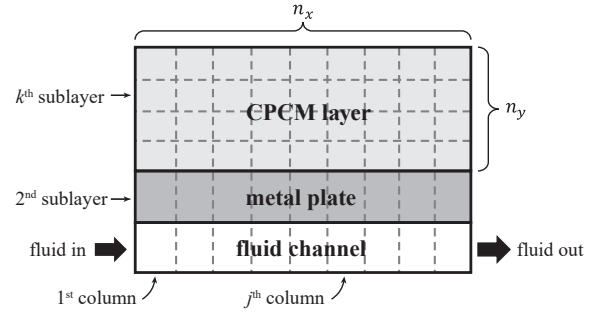


Figure 3. Schematic of the cross section of the TES, discretized into  $n_x$  columns and  $n_y + 2$  sublayers of control volumes.

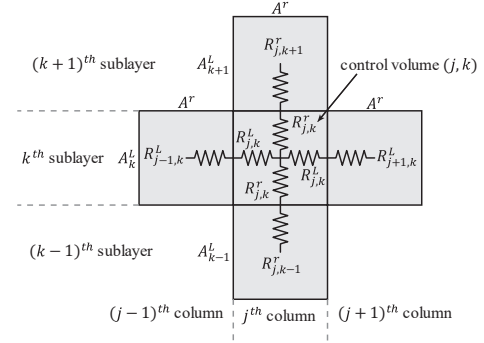


Figure 4. Thermal resistance network of control volume  $(j, k)$  in the CPCM layer and the four adjacent control volumes.

TES in the  $k^{\text{th}}$  sublayer ( $j = 1, 2, 3, \dots, n_x$  and  $k = 1, 2, 3, \dots, n_y + 2$ ). The  $\dot{Q}$  terms refer to the heat transfer rate from each of the four adjacent control volumes  $\{(j, k-1), (j, k+1), (j-1, k), (j+1, k)\}$  to control volume  $(j, k)$ , which are calculated in terms of the thermal resistances  $R$  and the temperature differences between each adjacent pair of control volumes.

$$\begin{aligned} m_k c_{p,(j,k)} \frac{dT_{j,k}}{dt} &= \dot{Q}_j^{(k-1) \rightarrow k} + \dot{Q}_j^{(k+1) \rightarrow k} + \dot{Q}_{j-1 \rightarrow j}^k + \dot{Q}_{j \rightarrow j+1}^k \\ &= \left[ \frac{T_{j,k-1} - T_{j,k}}{R_{j,k-1}^r + R_{j,k}^r} \right] + \left[ \frac{T_{j,k+1} - T_{j,k}}{R_{j,k+1}^r + R_{j,k}^r} \right] \\ &\quad + \left[ \frac{T_{j-1,k} - T_{j,k}}{R_{j-1,k}^L + R_{j,k}^L} \right] + \left[ \frac{T_{j+1,k} - T_{j,k}}{R_{j+1,k}^L + R_{j,k}^L} \right] \quad (9) \end{aligned}$$

Fig. 4 is a schematic of the thermal resistance network for a single control volume. In general, the thermal resistances in the vertical direction (superscript  $r$ ) are defined as  $R_{j,k}^r = \frac{h_k}{2k_{j,k}^r A^r}$ , and the thermal resistances in the horizontal direction (superscript  $L$ ) are defined as  $R_{j,k}^L = \frac{l}{2k_{j,k}^L A_k^L}$ , where  $h_k$  is the height of the  $k^{\text{th}}$  sublayer,  $l = L/n_x$  is the length of a single control volume,  $k_{j,k}^r$  and  $k_{j,k}^L$  are the thermal conductivities of control volume  $(j, k)$  in the vertical and horizontal directions, respectively, and  $A^r$  and  $A_k^L$  are the cross-sectional areas of the control volume in the vertical and horizontal directions,

respectively. The convective thermal resistance between the fluid control volumes and the metal plate control volumes is given as  $R_{j,k=1}^r = \frac{1}{\alpha_j A^r}$  where  $\alpha_j$  is the convective heat transfer coefficient in the  $j^{\text{th}}$  fluid control volume calculated using the Gnielinski correlation [15]. Conductive heat transfer between fluid control volumes is assumed to be negligible; all heat transfer between fluid control volumes is due to advection. In the control volumes at the boundaries of the TES (where  $j = 1, j = n_x, k = 1$ , or  $k = n_y + 2$ ), one or more heat transfer rate terms are removed in accordance with the adiabatic boundary condition.

Phase change is modeled using the sigmoid-based effective specific heat function described in [16] such that  $c_{p,(j,k)}$  is a function of the control volume temperature  $T_{j,k}$ . Near the PCM melting point,  $T_{melt}$ , the specific heat is greatly increased so that the latent heat of fusion is modeled as an equivalent increase in sensible heat over a small temperature interval around  $T_{melt}$ . These equations are combined and simplified to form the nonlinear state space model given in (10). The vector  $\mathbf{T}_{TES}$  contains the temperatures of each of the  $N_{TES} = n_x \times (n_y + 2)$  control volumes in the TES model, and the inlet temperature  $T_{TES,in}$  applies to the fluid control volume  $(j,k) = (1,1)$ , replacing  $T_{j=0,k=1}$ .

$$\dot{\mathbf{T}}_{TES} = A_{TES}\mathbf{T}_{TES} + B_{TES}T_{TES,in} \quad (10)$$

### C. Hybrid TMS Model

To simulate the hybrid TMS model, the fluid loop and TES models are combined using two coupling equations. Equation (11) relates the tank inlet temperature  $T_{t,in}$  to the TES outlet temperature and the bypass fluid temperature ( $T_{hx,f} = T_{TES,in}$ ). The matrix  $C_{TES} = [0 \dots 0 \quad f \quad 0 \dots 0]$  is a  $1 \times N_{TES}$  single-entry row vector, where the index of the only nonzero entry  $f$  is the same as the index of  $\mathbf{T}_{TES}$  corresponding to the temperature of the final fluid control volume, and  $f = \frac{\dot{m}_{TES}}{\dot{m}_p}$  is the fraction of the primary flow rate that passes through the TES. The scalar  $D_{TES} = 1 - f$  is the fraction of the primary flow rate that bypasses the TES. Equation (12) relates the TES inlet temperature,  $T_{TES,in}$ , to the heat exchanger outlet temperature. The matrix  $C_{fl} = [0 \quad 0 \quad 1 \quad 0 \quad 0]$  is the  $1 \times N_{fl}$  single-entry row vector where the index of the nonzero entry is the same as the index of  $\mathbf{T}_{fl}$  corresponding to the heat exchanger outlet temperature.

$$T_{t,in} = C_{TES}\mathbf{T}_{TES} + D_{TES}T_{TES,in} \quad (11)$$

$$T_{TES,in} = C_{fl}\mathbf{T}_{fl} \quad (12)$$

Combining (7), (10), (11), and (12) yields the hybrid TMS state space model given by (13).

$$\begin{bmatrix} \dot{\mathbf{T}}_{fl} \\ \dot{\mathbf{T}}_{TES} \end{bmatrix} = \begin{bmatrix} A_{fl} + B_t D_{TES} C_{fl} & B_t C_{TES} \\ B_{TES} C_{fl} & A_{TES} \end{bmatrix} \begin{bmatrix} \mathbf{T}_{fl} \\ \mathbf{T}_{TES} \end{bmatrix} + \begin{bmatrix} B_{fl} \\ 0 \end{bmatrix} \dot{Q} \quad (13)$$

## III. ESTIMATION AND CONTROL

This section defines the state of charge of the TES device and discusses how it is estimated using the model states. Then the control objectives are defined, followed by a description of the proposed logic-based controller for the hybrid TMS.

### A. State of Charge Estimation

The state of charge of the TES device,  $x_{SOC}$ , is defined in (14) as a linear function of enthalpy,  $Q$ . The maximum  $x_{SOC}$  value of 1 corresponds to a minimum enthalpy stored in the TES when the PCM temperature is a uniform  $T_{min}$ , and the minimum  $x_{SOC}$  value of 0 corresponds to a maximum enthalpy stored in the TES when the PCM temperature is a uniform  $T_{max}$ . The temperatures  $T_{min}$  and  $T_{max}$  are predefined constants denoting the minimum and maximum expected fluid temperatures. Decreasing  $x_{SOC}$ , or increasing enthalpy, is referred to as *discharging* the TES, and likewise increasing  $x_{SOC}$ , or decreasing enthalpy, is referred to as *charging* or *recharging* the TES.

$$x_{soc}(t) = \begin{cases} 1: & Q(t) < Q_{min} \\ \frac{Q_{max} - Q(t)}{Q_{max} - Q_{min}}: & Q_{min} \leq Q(t) \leq Q_{max} \\ 0: & Q(t) > Q_{max} \end{cases} \quad (14)$$

With all the internal states in the TES known to the controller,  $Q$  can be calculated as a function of temperature using the effective specific heat function described in [16].

### B. Controller Objectives

A controller that manages a hybrid TMS without knowledge of upcoming heat loads must balance three objectives.

- 1) Objective 1 requires that the TES be recharged whenever possible to conserve its state of charge. In other words, since the duration and magnitude of the incoming heat loads are unknown, the TES must maintain as high of a state of charge as possible until it is needed. Recharging is possible whenever the fluid temperature leaving the heat exchanger is less than the plate surface temperature in the TES.
- 2) Objective 2 is to maintain the temperature at the surface of the cold plate,  $T_{cp,w}$ , below some upper limit,  $\bar{T}_{cp,obj}$ . This objective can be achieved by discharging the TES, further cooling the fluid before it enters the cold plate. However, the TES should be discharged at the minimum heat transfer rate that still meets the cold plate surface temperature objective to minimize the decrease in state of charge. Discharging heat rates tend to be larger in magnitude than recharging heat rates due to the low melting point of the PCM, so  $x_{SOC}$  can decrease quickly during a discharge but takes longer to recover during a recharge.
- 3) Objective 3 is to minimize the amount of heat entering the primary fluid loop through the heat exchanger. While uncommon, in the event that the chiller fluid temperature is higher than the primary fluid temperature, the primary flow rate should be reduced to avoid heat transfer into the cooling loop.

### C. Proposed Heuristic Control Strategy

The proposed heuristic, or logic-based, controller is outlined in the flowchart shown in Fig. 5. The flowchart is separated

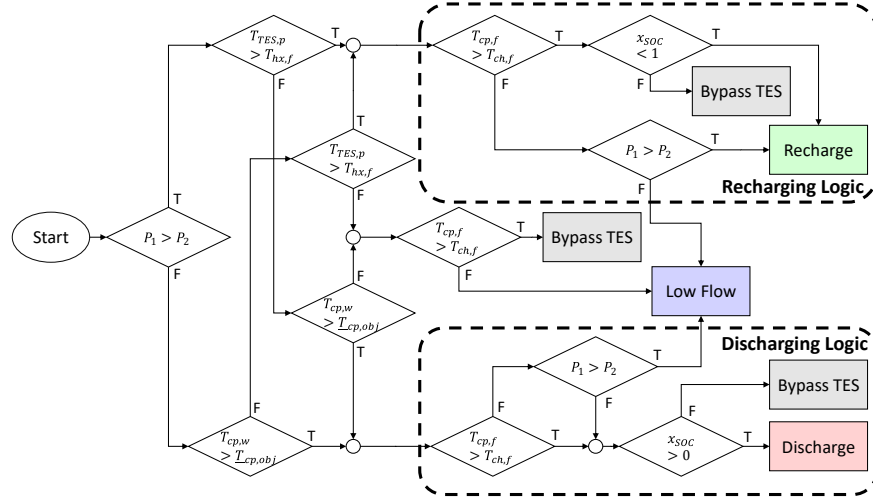


Figure 5. Flowchart for the proposed heuristic control strategy. After deciding whether recharging is possible or whether discharging is necessary, the controller chooses one of four modes of operation: recharge the TES, discharge the TES, bypass the TES, or bypass the TES and reduce the primary flow rate to low.

Table II  
CONTROL ACTIONS DEFINED FOR EACH MODE

| Operation Mode        | Control Action                                                                        | Control Input Equation                                                                                                                                                                                           |
|-----------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recharge              | Maximum primary flow rate;<br>Maximum TES flow rate                                   | $\dot{m}_p = \dot{m}_{p,max}$<br>$\dot{m}_{TES} = \dot{m}_{p,max}$                                                                                                                                               |
| Discharge             | Maximum primary flow rate;<br>TES flow rate proportional to<br>cold plate temperature | $\dot{m}_p = \dot{m}_{p,max}$<br>$\dot{m}_{TES} = \dot{m}_{p,max} \left( \frac{T_{cp,w} - \underline{T}_{cp,obj}}{\bar{T}_{cp,obj} - \underline{T}_{cp,obj}} \right); 0 \leq \dot{m}_{TES} \leq \dot{m}_{p,max}$ |
| Bypass TES            | Maximum primary flow rate;<br>Zero flow through TES                                   | $\dot{m}_p = \dot{m}_{p,max}$<br>$\dot{m}_{TES} = 0$                                                                                                                                                             |
| Low Primary Flow Rate | Reduce primary flow rate to<br>10% of maximum flow rate;<br>Zero flow through TES     | $\dot{m}_p = 0.1\dot{m}_{p,max}$<br>$\dot{m}_{TES} = 0$                                                                                                                                                          |

into two sections: logic for recharging and logic for discharging. The initial logic statements determine if 1) recharging is possible or 2) discharging is necessary. The controller uses temperature and state of charge feedback to decide between one of four operation modes: recharging, discharging, bypassing the TES, and reducing the primary flow rate to a low flow condition. Table II outlines how the control input variables  $\dot{m}_p$  and  $\dot{m}_{TES}$  are calculated for each mode. The primary fluid mass flow rate,  $\dot{m}_p$ , can vary between  $\dot{m}_{p,min} = 6.3$  g/s and  $\dot{m}_{p,max} = 63$  g/s, and the TES flow rate can vary from 0 to  $\dot{m}_p$ . The controller updates the operation mode and control inputs every 0.25 seconds. To meet Objective 2, two threshold variables are defined:  $\bar{T}_{cp,obj} = 45^\circ\text{C}$ , the upper temperature limit at which the maximum rate of discharge of the TES is necessary, and  $\underline{T}_{cp,obj} = 35^\circ\text{C}$ , the lower temperature limit below which discharging is not necessary to maintain Objective 2.

The variables  $P_1$  and  $P_2$ , defined in (15) and (16), respectively, such that  $P_1, P_2 \in \{0, 1, \dots, 10\}$ , quantify the priorities of recharging the TES (Objective 1) and maintaining

$T_{cp,w}$  (Objective 2), respectively. They are used to determine whether the controller should conserve state of charge or discharge the TES to reduce the cold plate surface temperature when the two objectives are in conflict. The variable  $P_1$  is defined as a function of  $x_{SOC}$  such that  $P_1$  increases as  $x_{SOC}$  decreases; in other words, the priority of recharging the TES increases as its state of charge decreases. The variable  $P_2$  is a linear function of  $T_{cp,w}$  for  $\underline{T}_{cp,obj} \leq T_{cp,w} \leq \bar{T}_{cp,obj}$ , increasing from 0 to 6 as  $T_{cp,w}$  increases from  $\underline{T}_{cp,obj}$  to  $\bar{T}_{cp,obj}$ , so that the priority of maintaining  $T_{cp,w}$  increases as  $T_{cp,w}$  nears its upper limit. If  $T_{cp,w}$  exceeds the upper limit  $\bar{T}_{cp,obj}$ ,  $P_2 = 10$ , its maximum value. For  $T_{cp,w}$  below the lower limit  $\underline{T}_{cp,obj}$ ,  $P_2 = 0$ .

$$P_1 = \text{floor}[10 \times (1 - x_{SOC})] \quad (15)$$

$$P_2 = \begin{cases} 0: & T_{cp,w} < \underline{T}_{cp,obj} \\ \text{floor}\left[6 \times \frac{T_{cp,w} - \underline{T}_{cp,obj}}{\bar{T}_{cp,obj} - \underline{T}_{cp,obj}}\right]: & \underline{T}_{cp,obj} \leq T_{cp,w} < \bar{T}_{cp,obj} \\ 10: & T_{cp,w} \geq \bar{T}_{cp,obj} \end{cases} \quad (16)$$



The flowchart in Fig. 5 first determines if recharging the TES is possible or if discharging is necessary to reduce the cold plate surface temperature. Recharging the TES is possible when the TES plate temperature,  $T_{TES,p}$ , is greater than the temperature of the primary fluid leaving the heat exchanger,  $T_{hx,f}$ . Discharging the TES is necessary when  $T_{cp,w} > \bar{T}_{cp,obj}$ . These two conditions are not mutually exclusive, so comparing  $P_1$  and  $P_2$  determines which objective takes precedence, and the controller continues into either the recharging logic section or the discharging logic section. If neither condition is met, the controller checks whether to address Objective 3, which requires that the primary flow rate be reduced to its low value if  $T_{ch,f} > T_{cp,f}$  to prevent too much heat from entering the primary fluid loop through the heat exchanger. Otherwise, the controller bypasses the TES without reducing the primary flow rate.

The recharging logic section of Fig. 5 determines whether to recharge the TES to meet Objective 1, assuming recharging is possible. The controller first checks whether the flow rate should be reduced to meet Objective 3, but since recharging is possible, this objective can be ignored if recharging the TES (Objective 1) has a higher priority than maintaining the cold plate temperature (Objective 2), or  $P_1 > P_2$ . If Objective 3 does not need to be addressed, the controller will send the system into recharging mode by maximizing the flow rate through the TES, provided  $x_{SOC} < 1$ .

Finally, the discharging logic section in Fig. 5 determines whether to discharge the TES, assuming discharging is necessary. Again, the controller checks whether  $T_{ch,f} > T_{cp,f}$  to decide whether to address Objective 3 and will choose the low flow mode if conserving state of charge has a higher priority than maintaining the cold plate wall temperature ( $P_1 > P_2$ ). Otherwise, the controller will discharge the TES at a flow rate proportional to the cold plate temperature, provided  $x_{SOC} > 0$ .

#### IV. RESULTS AND DISCUSSION

The following case study demonstrates the proposed heuristic controller. The cold plate heater applies a random sequence of heat pulses (varying in magnitude between 0 and 6 kW) while the chiller fluid temperature varies between 5°C and 25°C for a total simulation time of 250 seconds. The hybrid TMS controller attempts to keep the cold plate surface temperature  $T_{cp,w}$  below  $\bar{T}_{cp,obj} = 45^\circ\text{C}$ , and it will begin to discharge the TES when  $T_{cp,w}$  exceeds  $\underline{T}_{cp,obj} = 35^\circ\text{C}$ . The mass of fluid in the tank is 500 g. The performance of the hybrid TMS will then be compared to the performance of a conventional TMS without integrated TES.

##### A. Heuristic Controller Demonstration

In the figures that follow, shading corresponds to the operation modes the heuristic controller chooses from—red shading indicates discharging periods, green shading indicates recharging periods, blue shading indicates periods of low primary flow rate, and no shading indicates that the TES is bypassed. Fig. 6 shows each of the heat transfer rates

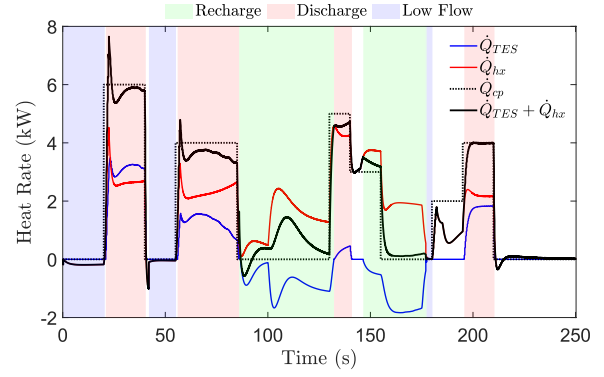


Figure 6. Heat transfer rates for the hybrid TMS. Whenever the total heat rejection rate ( $\dot{Q}_{TES} + \dot{Q}_{hx}$ ; solid black line) is less than the heat load ( $\dot{Q}_{cp}$ ; dotted black line), more heat is entering the system than leaving, and the fluid temperatures increase. When the TES is recharging,  $\dot{Q}_{TES} + \dot{Q}_{hx}$  can be negative because the heat leaving the TES enters the fluid before being rejected by the heat exchanger.

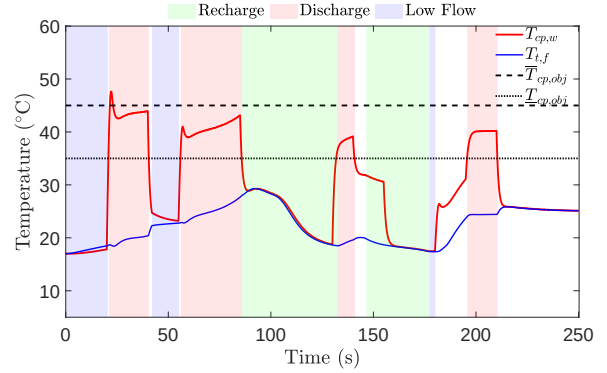


Figure 7. The fluid and surface temperatures in the cold plate determine  $P_2$ , the priority of discharging the TES. When  $T_{cp,w}$  (red line) is below the lower temperature limit  $\underline{T}_{cp,obj} = 35^\circ\text{C}$  (dotted line),  $P_2 = 0$  and discharging does not occur. The controller will always discharge the TES if  $T_{cp,w}$  exceeds the upper temperature limit  $\bar{T}_{cp,obj} = 45^\circ\text{C}$  (dashed line), unless  $x_{SOC} = 0$ , because in that case  $P_2 = 10$ , the highest priority.

into or out of the primary fluid loop over the 250 second simulation. The disturbance heat load,  $\dot{Q}_{cp}$ , depicted by the dotted line in Fig. 6, is applied as a series of pulses of random magnitude and duration. The hybrid TMS responds to each heat pulse by rejecting heat through the heat exchanger and storing heat in the TES. Any heat that is not rejected or absorbed is stored as sensible heat in the tank, increasing the fluid temperatures in the system. As expected, the total heat rejection rate ( $\dot{Q}_{TES} + \dot{Q}_{hx}$ ; solid black line) tends to follow the heat load. When the heat exchanger cannot reject the entire heat load, the controller discharges the TES to increase the cooling capacity of the overall system.

The main purpose of the TMS is to keep the cold plate surface temperature below its upper limit of  $45^\circ\text{C}$ . Fig. 7 shows that the hybrid TMS maintains this objective, except for a brief overshoot at  $t = 25$  s when the 6 kW heat load is applied. Note that  $\dot{Q}_{cp}$  is approximately proportional to the temperature difference between the cold plate surface ( $T_{cp,w}$ ;

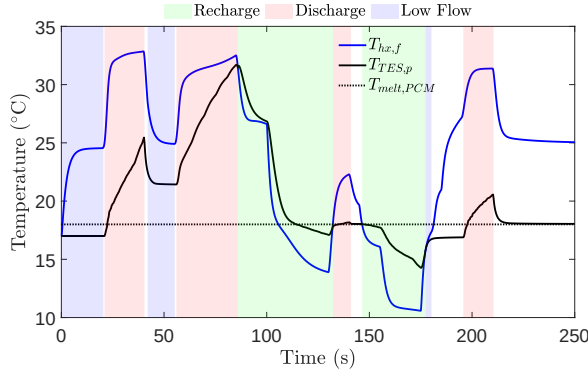


Figure 8. The temperature of the primary fluid leaving the heat exchanger ( $T_{hx,f}$ ; blue line) and the mean temperature of the TES metal plate ( $T_{TES,p}$ ; black line) determine the conditions required for recharging the TES.

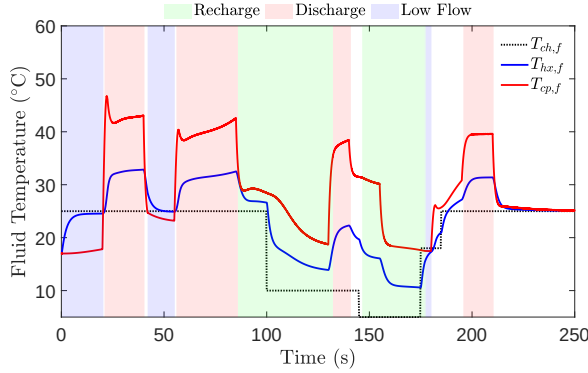


Figure 9. The heat exchanger inlet and outlet primary fluid temperatures ( $T_{cp,f}$ ; red line and  $T_{hx,f}$ ; blue line, respectively) and the chiller fluid temperature ( $T_{ch,f}$ ; dotted line) demonstrate the purpose of the low flow rate mode. In the blue shaded regions, the primary fluid temperature increases across the heat exchanger, and so the rate of heat transfer into the primary fluid loop through the heat exchanger must be minimized.

red line), and the incoming fluid from the tank ( $T_{t,f}$ ; blue line).

Next, Fig. 8 highlights how the heuristic controller switches into the recharging mode, depicted using green shading, based on estimates of the TES plate temperature ( $T_{TES,p}$ ; black line) and the heat exchanger primary fluid outlet temperature ( $T_{hx,f}$ ; blue line). More specifically, the system enters recharging mode only when  $T_{TES,p} > T_{hx,f}$ .

While the system can enter discharge mode whenever  $T_{TES,p} < T_{hx,f}$ , Fig. 8 and Fig. 9 show that competing objectives can prevent the system from doing so. For example, when the chiller fluid temperature ( $T_{ch,f}$ ; dotted line in Fig. 9) is higher than the temperature of the primary fluid entering the heat exchanger ( $T_{cp,f}$ ; red line in Fig. 9), such as from  $t = 0$  s to  $t = 21$  or between  $t = 42$  s and  $t = 55$  s, the controller enters the low flow rate mode to minimize the heat entering the primary fluid loop through the heat exchanger (from the secondary fluid). During these periods, the temperature of the primary fluid leaving the heat exchanger ( $T_{hx,f}$ ; blue line in 9) is greater than the temperature of the primary fluid entering the heat exchanger, indicating that heat is still entering the system

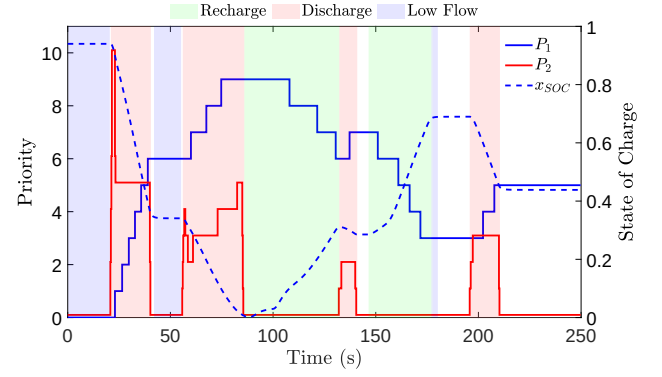


Figure 10. Priorities of recharging the TES ( $P_1$ ; solid blue line) and maintaining the cold plate temperature ( $P_2$ ; solid red line) on the left y-axis, and the state of charge ( $x_{SOC}$ ; dashed blue line) on the right y-axis.

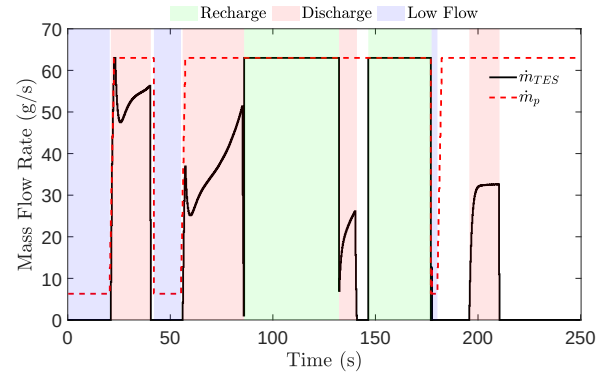


Figure 11. Primary and TES flow rates ( $\dot{m}_p$  and  $\dot{m}_{TES}$ , respectively). The primary flow rate only decreases during the low flow mode, but the TES flow rate is determined based on the cold plate temperature when discharging.

at a minimal rate. These plots highlight the complexity of the decision-making executed by the heuristic controller.

Fig. 10 shows the controller's priorities and the calculated state of charge—specifically, how  $P_1$  changes in response to the state of charge, which decreases during the discharging periods and increases during the recharging periods. Because all the state temperatures are initialized at a temperature of 17°C,  $x_{SOC}$  is not initially equal to 1. A state of charge of 1 means that the average temperature inside the TES is uniformly  $T_{min} = 5^\circ\text{C}$ . Fig. 10 also shows that  $x_{SOC}$  tends to decrease during a discharge period at a much faster rate than it increases during a recharge period, demonstrating why it is important to conserve state of charge. This is particularly true from  $t = 55$  s to  $t = 83$  s when the state of charge drops to zero. If the TES had been discharged at a faster rate, the state of charge would have been depleted before the end of the heat pulse, possibly leading to the cold plate overheating. Since the SOC decreases quickly during the first 100 seconds of the simulation, the priority of recharging is almost always higher than the priority of maintaining the cold plate temperature.

Fig. 11 shows the primary and TES mass flow rates, i.e. the control inputs calculated by the controller based on the chosen operation mode. During the low flow mode, the primary flow



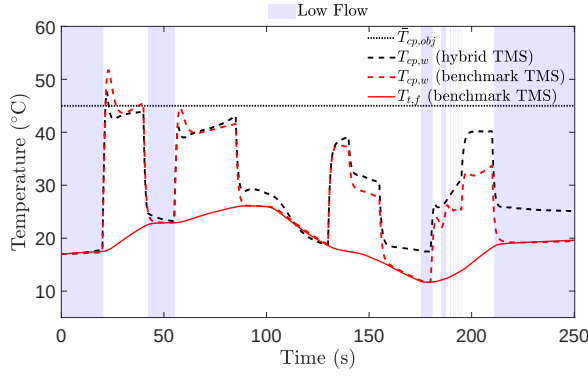


Figure 12. Surface temperatures in the cold plate for the benchmark system (dashed red line) compared to the hybrid system (dashed black line). Also shown is the tank temperature in the benchmark TMS (solid red line).

rate drops to 10% of its maximum value. The TES flow rate is always maximized during recharging mode so as to maximize the rate of recharge, and it varies depending on the cold plate surface temperature during the discharging mode to limit the rate of discharge.

### B. Comparison with a Benchmark System

This section compares a benchmark system—defined as a conventional TMS without an integrated TES device—to the hybrid TMS operating with the proposed heuristic controller. The controller for the benchmark TMS always uses the maximum primary flow rate ( $m_p = 63$  g/s), except if  $T_{ch,f} > T_{cp,f}$ , in which case it will use the low primary flow rate mode. Given that one of the advantages of a hybrid TMS is the ability to downsize the components, the sizes of the heat exchanger and the mass of fluid in the tank in the benchmark system are increased such that both systems achieve similar results in terms of cold plate temperature regulation. A process of iteratively increasing the heat exchanger mass and surface area and the tank fluid mass, then comparing the simulation results to those of the hybrid TMS determines that the benchmark TMS requires a heat exchanger with a fluid flow length four times that of the hybrid TMS and a tank fluid volume also four times that of the hybrid TMS. Fig. 12 compares the cold plate wall temperature achieved using the benchmark TMS (dashed red line) with that of the hybrid TMS (dashed black line) for the heat load profile originally shown in Fig. 6.

Next, the heat rejection rates achieved using each system are compared in Fig. 13. The heat rejection rate of the hybrid TMS (solid black line) more closely tracks the heat addition rate of the cold plate (dotted line) than does the benchmark TMS (dashed red line), despite the smaller heat exchanger in the hybrid TMS. This difference is due to the fast dynamics of the TES, which enable it to absorb heat quickly, while the heat exchanger has a slower transient response.

The most significant impact of the hybrid TMS can be seen by comparing the tank inlet temperatures for both the hybrid TMS and the benchmark system, as shown in Fig. 14. In the hybrid TMS, the TES regulates the cold plate surface

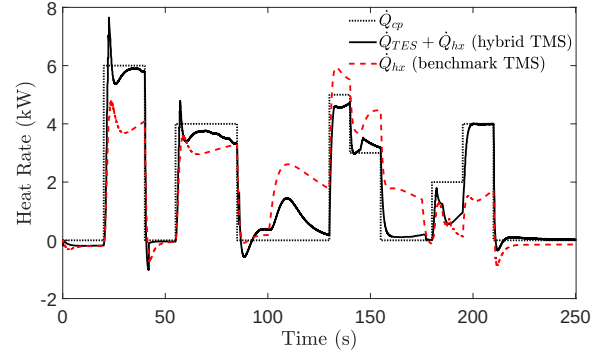


Figure 13. Heat rejection rates compared between the benchmark TMS (dashed red line) and the hybrid TMS (solid black line). Note that for the benchmark TMS,  $\dot{Q}_{TES} = 0$ , so  $\dot{Q}_{hx}$  is the total heat rejection rate. Also shown is the heat addition rate ( $\dot{Q}_{cp}$ ; dotted line), which is the same in both case studies.

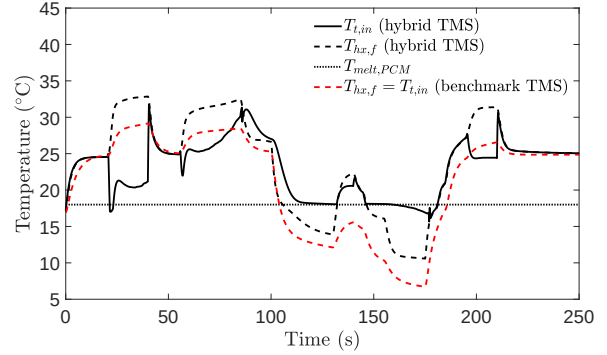


Figure 14. Comparison between the temperature of the fluid entering the tank,  $T_{t,in}$ , in both the hybrid TMS (solid black line) and the benchmark TMS (dashed red line). In the hybrid TMS,  $T_{t,in}$  is a function of both the temperature of the fluid leaving the heat exchanger ( $T_{hx,f}$ ; dashed black line), and the temperature of the fluid leaving the TES (not shown), but in the benchmark TMS,  $T_{t,in} = T_{hx,f}$ .

temperature by decreasing the temperature of the fluid entering the tank ( $T_{t,in}$ ; solid black line), which in turn decreases the temperature of the fluid entering the cold plate,  $T_{t,f}$ . By precooling the TES,  $T_{t,in}$  can even be reduced below the temperature of the heat exchanger's secondary fluid (which is 25°C for the first 100 seconds as shown in Fig. 9), which is not possible with the benchmark TMS—the period between  $t = 20$  s and  $t = 40$  s in Fig. 14 shows a clear example of the effect of precooling. Instead,  $T_{t,in}$  in the benchmark system (dashed red line) increases during a heat pulse because the heat exchanger alone cannot reject the full heat load instantaneously because of its slow transient response. The temperatures of the fluid entering the tank for both the hybrid and benchmark TMSs demonstrate the advantage of using the TES to further decrease the fluid temperature when the heat exchanger cannot reject the entire heat load. The TES can reduce  $T_{t,in}$  by over 10°C, which is necessary during the initial 6 kW heat load between  $t = 20$  and  $t = 40$  seconds. The ability to store extra heat rejection capacity is a key advantage of a hybrid TMS.

During recharge periods, such as between  $t = 82$  s and  $t = 130$  s in Fig. 14,  $T_{t,in}$  is higher in the hybrid TMS than in the benchmark TMS because the recharging TES is a heat source rather than a heat sink. This demonstrates the TES's ability to distribute a transient heat load over a longer time period while reducing the peak load experienced by the heat exchanger. Note that in Fig. 14,  $T_{t,in}$  in the hybrid TMS tends to stagnate around the melting temperature of the PCM ( $T_{melt,PCM}$ ; dotted line) because while the TES is recharging, the temperature of the PCM control volumes decrease more slowly while undergoing phase change.

## V. CONCLUSION

This paper presents a heuristic logic-based controller for a hybrid thermal management system (TMS) with the goal of regulating the temperature at the surface of a cold plate heater, subjected to a randomly varying and unknown thermal load. The heuristic controller balances three objectives: 1) to recharge the TES whenever possible, 2) to maintain the surface temperature of a cold plate below  $45^{\circ}\text{C}$  while conserving state of charge to avoid premature depletion of storage capacity, and 3) to minimize the amount of heat entering the primary fluid loop from the secondary fluid through the heat exchanger. Based on an estimate of the state of charge of the TES device and knowledge of the current fluid loop temperatures, the controller selects one of four operating modes: recharge the TES, discharge the TES, bypass the TES, or reduce the primary flow rate to 10% of its maximum value.

The performance of the controller is demonstrated through a simulated case study in which the hybrid TMS is subjected to randomly varying heat pulses up to 6 kW and a time-varying secondary fluid temperature. As the temperature of the cold plate approaches  $45^{\circ}\text{C}$ , the controller sends more fluid through the TES, reducing the temperature of the fluid entering the cold plate while depleting the state of charge more quickly. During the 250 second simulation, the cold plate temperature only exceeds the  $45^{\circ}\text{C}$  upper limit for a period of two seconds during the largest heat pulse.

In a second case study, the advantages of the hybrid TMS architecture over a conventional (benchmark) TMS that does not utilize an integrated TES device are demonstrated. The benchmark TMS utilizes a similar controller that toggles between high flow rate and low flow rate modes. However, in order for the benchmark TMS to keep the cold plate temperature below the same upper bound of  $45^{\circ}\text{C}$ , it needs both a heat exchanger and fluid tank four times larger than those included in the hybrid TMS. The hybrid TMS is able to operate with smaller components by leveraging the fast dynamics of the TES and the low melting point of the PCM to precool the primary fluid below the temperature of the secondary fluid in the heat exchanger. Moreover, even with larger components, the cold plate temperature in the benchmark system exceeds the upper limit during the largest heat pulse. Future work will include experimental implementation and validation of the proposed control strategy, as well as consideration of advanced

control strategies that leverage some knowledge of future heat loads.

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